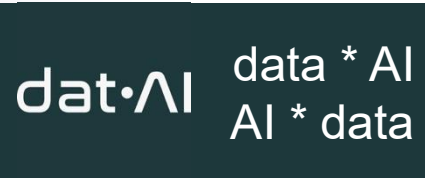


Matteo Palmonari

matteo.palmonari@unimib.it

Data Semantics: Introduction



Just to warm up #1

□ Matteo Palmonari

- AI, Knowledge Representation, Data Integration
- How to represent? How to match representations?
How to learn representations?
- Information Extraction and Data Enrichment



□ Riccardo Pozzi

- Deep Learning, Knowledge Graphs, NLP



Course Schedule

- Tuesday 9.00 - 11.30
- Wednesday 9.00 - 10.30
- Hands-on lessons mixed into the course schedule
- Backup links:
 - ✕ Matteo Palmonari's webex room:
unimib.webex.com/meet/matteo.palmonari
 - ✕ Riccardo Pozzi's webex room:
unimib.webex.com/meet/riccardo.pozzi

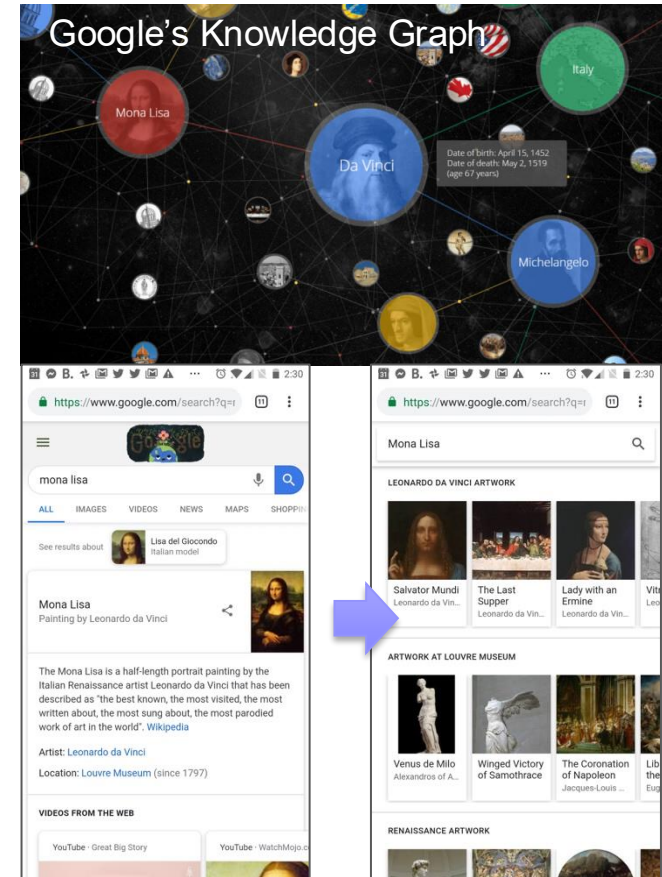
Just to warm up #2

- What is your background (BA)?
 - Speak up

- What do you think data semantics is about?
 - Speak up

Data Semantics / What (the intuition)

- What is data semantics?
 - Data semantics is about
 - **modeling the meaning** of data and using it to improve their interpretation, interoperability across systems, and subsequent processing
 - How can we model data better so that ...
 - **interpreting the meaning** of data and use this imputed interpretation to improve data processing
 - How can we interpret data better so that ...



Data Semantics / Why

■ When should we take care of data semantics?

- When we need to **integrate** different and heterogeneous data sources
- When we need to **make a dataset accessible** to parties who do not know the dataset
- When we need to **access** a dataset that is produced by third parties
- When we need to **infer meaning of data**

■ Why?

- To **improve data processing** in crucial steps of the data science lifecycle
- **Data preparation** (including: cleaning, integration, transformation) can cover up to 80% of effort in a data science project

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Big Data Analytics and Data Semantics (Volume Velocity **Variety**)

2020 *This Is What Happens In An Internet Minute*

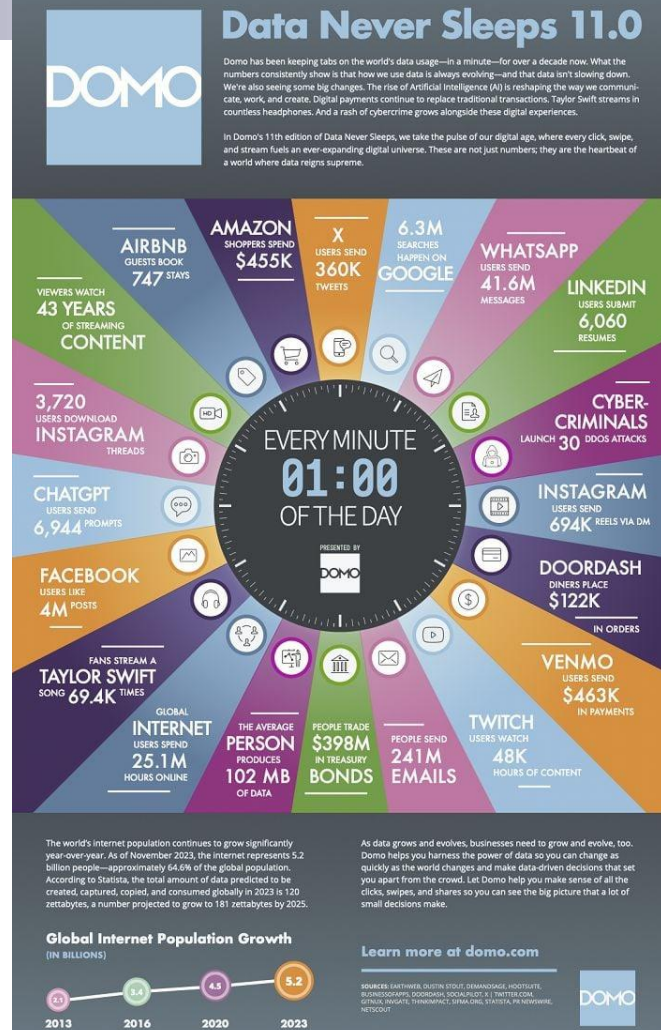


2021 *This Is What Happens In An Internet Minute*



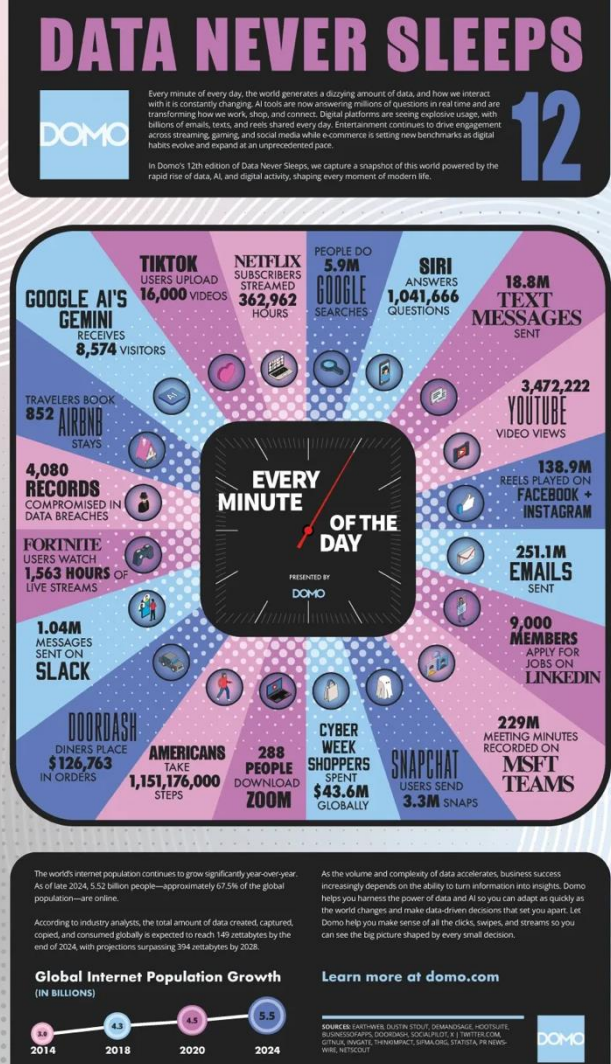
2023

- Texts (incl. prompts)
- Images
- Videos
- Also a lot of relational data (interactions, transactions, etc.)



2024

- More ChatBots
- More interactions
- ...



2025

...

More AI

DATA NEVER SLEEPS



Every minute of every day, the world generates a dizzying amount of data, and how we interact with it is constantly changing. AI tools are now answering millions of questions in real time and are transforming how we work, shop, and connect. Digital platforms are seeing explosive usage, with billions of emails, texts, and reels shared every day. Entertainment continues to drive engagement across streaming, gaming, and social media while e-commerce is setting new benchmarks as digital habits evolve and expand at an unprecedented pace.

In Domo's 12th edition of Data Never Sleeps, we capture a snapshot of this world powered by the rapid rise of data, AI, and digital activity, shaping every moment of modern life.

12



The world's internet population continues to grow significantly year-over-year. As of late 2024, 5.5 billion people—approximately 67.5% of the global population—are online.

According to industry analysts, the total amount of data created, captured, copied, and consumed globally is expected to reach 149 zettabytes by the end of 2024, with projections surpassing 394 zettabytes by 2028.

Global Internet Population Growth (IN BILLIONS)



As the volume and complexity of data accelerates, business success increasingly depends on the ability to turn information into insights. Domo helps you harness the power of data and AI so you can adapt as quickly as the world changes and make data-driven decisions that set you apart. Let Domo help you make sense of all the clicks, swipes, and streams so you can see the big picture shaped by every small decision.

Learn more at domo.com

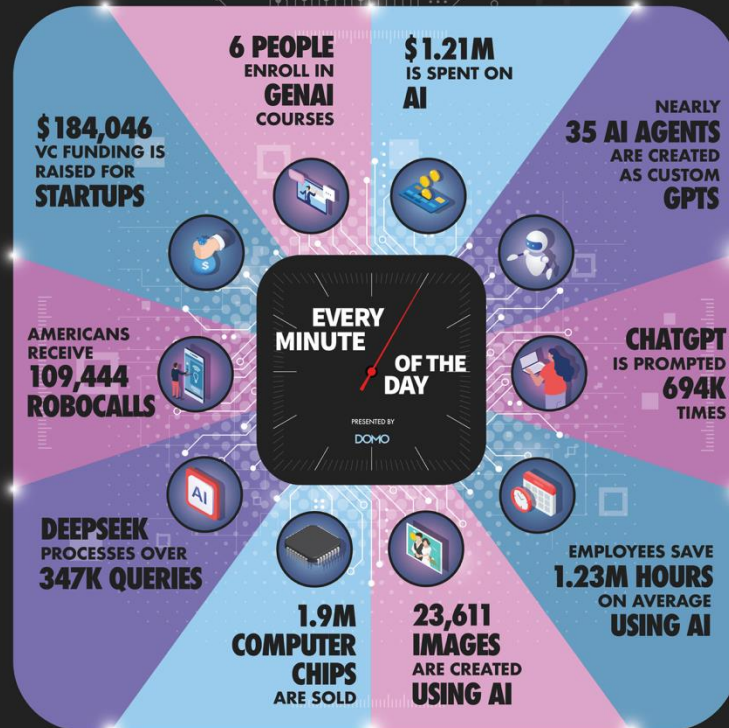
SOURCES: LAMARCA, DUSTIN SYDUT, DEMARKSAGE, HEDYBITE, BUSINESSINSIDER, DODGSON, SCHWARTZ & PARTNERS, CNET, DIGITAL, RANGATE, THUNDERBOLT, SPINAL DIAL, STATISTA, PR NEWS, WIRE, NETSCOUT



DATA NEVER SLEEPS AI EDITION 2025



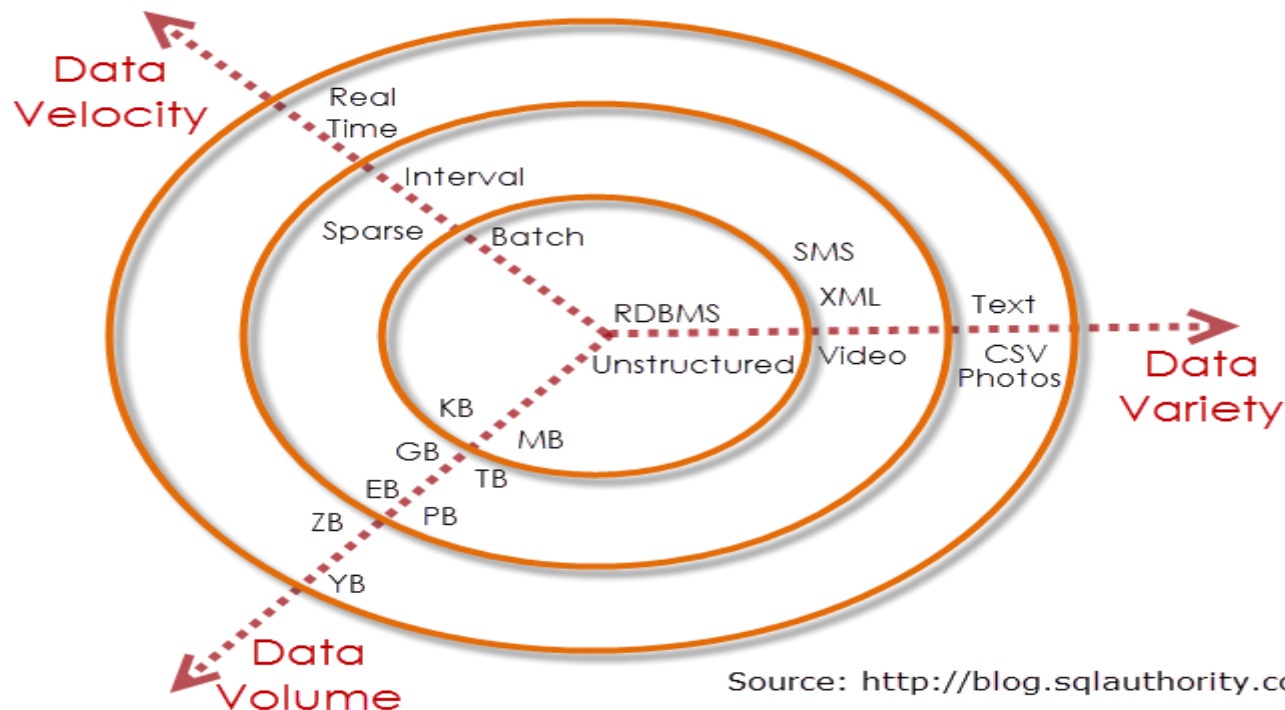
Domo has tracked minute-by-minute digital engagement for over a decade, and AI is moving from experimental to essential at unprecedented speed. Spending on artificial intelligence has more than tripled since 2024. From boardrooms to living rooms, this special second Data Never Sleeps AI edition reveals how AI is reshaping the global economy, 60 seconds at a time.



SOURCES: COINTEL, GADGET, COURIERA, CROPIK, ENTREPRENEUR, EVERPOINT, GOLD BOX COMPANIES, OPENAI, PRECEDENCE RESEARCH, S&P, TRADING ECONOMICS

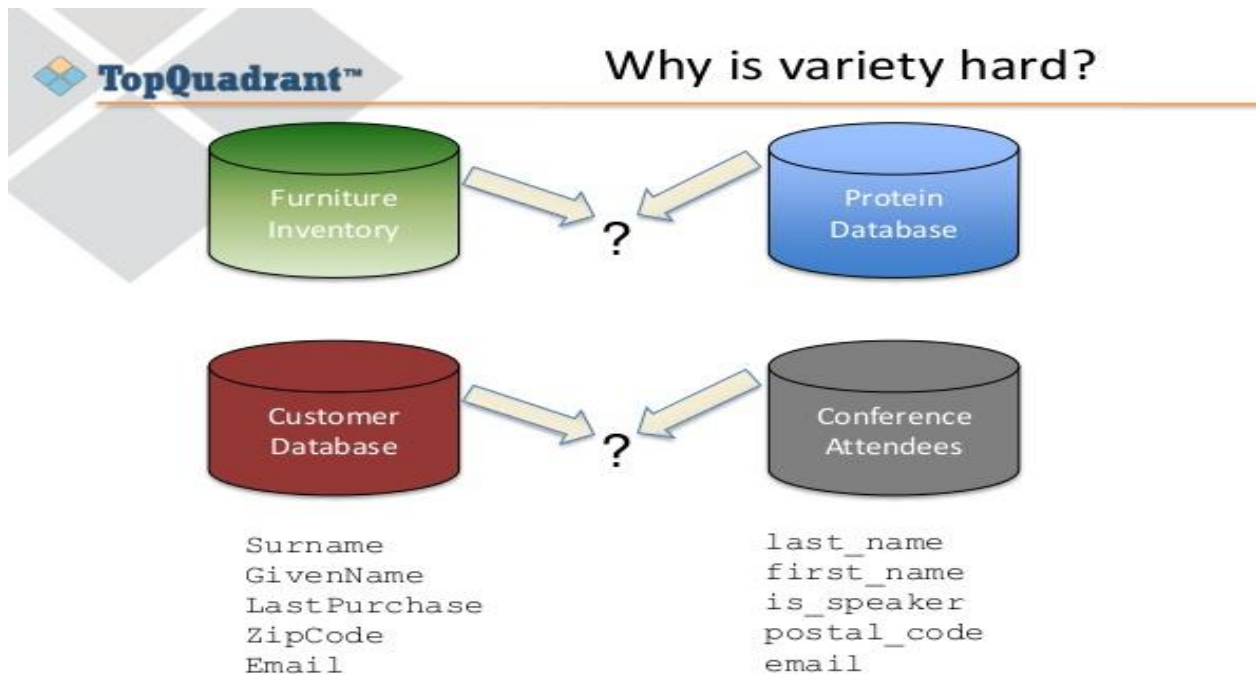
Big Data: Volume Velocity Variety

3Vs of Big Data



Source: <http://blog.sqlauthority.com>

Variety & Semantics (First Dive)



Heterogeneity of Representations

R(FirstName, LastName, Region, State)

Patrick	Metzisi	MM	Kenia
---------	---------	----	-------

Patrick	Metzisi	Masai Mara	KE
---------	---------	------------	----

(a) Two tuples

R1(FirstName, LastName, Region)

Patrick	Metzisi	MM
---------	---------	----

R2(Region, State)

MM	Kenia
----	-------

R3(State, Continent)

Kenia	Africa
-------	--------

Patrick	Mezisi	Masai Mara
---------	--------	------------

Masai Mara	KE
------------	----

KE	Africa
----	--------

(b) Two hierarchical groups of tuples

```
<country>
  <name> Kenya </name>
  <cities> Nairobi, Mombasa, Malindi
</cities>
  <lakes>
    <name> Lake Victoria </name>
  </lakes>
</country>
```

```
<country>
  Kenya
  <city> Nairobi </city>
  <city> Mombasa </city>
  <lakes>
    <lake> Lake Victoria </lake>
  </lakes>
</country>
```

(c) Two XML records

Heterogeneity of Representations

Heterogenous sources in different formats
(RDB, ODB, XML, non structured data, multimedia, ...)

Different representations

```
<country>
Kenia
<city> Nairobi </city>
<city> Mombasa </city>
<lakes>
<lake> Lake Victoria </lake>
</lakes>
</country>
```



Google search results for Nairobi, Kenya. The page shows the Wikipedia article for Nairobi, including a photo of a giraffe and a city skyline. The article text describes Nairobi as the capital of Kenya, known for its climate and tourism. It mentions the population and its status as a major city in East Africa.

R1(FirstName, LastName, Region)

Patrick	Metzisi	MM
---------	---------	----

R2(Region, State)

MM	Kenia
----	-------

R3(State, Continent)

Kenia	Africa
-------	--------

Heterogeneity of Representations

Heterogenous sources in different formats
(semi-structured data, e.g., web tables)

No	City/ Town	Status	Population	County	Ref
	Nairobi Metro	Metro	6,547,547	Nairobi Kajiado Machakos Kiambu	[1]
1.	 Nairobi	City	4,394,073	 Nairobi	[2]
2.	 Mombasa	City	1,208,333	 Mombasa	
3.	Kisumu	City	721,082	Kisumu	
4.	Nakuru	Municipality	570,674	Nakuru	
5.	Ruiru	Municipality	490,120	Kiambu	
6.	Eldoret	Municipality	475,716	Uasin Gishu	
7.	Kikuyu	Municipality	323,881	Kiambu	
8.	Thika	Municipality	251,407	Kiambu/Murang'a	
9.	Meru	Municipality	240,900	Meru	
10.	Kangundo-Tala	Town Council	218,557	Machakos	
11.	Malindi	Municipality	207,253	Kilifi	
12.	Naivasha	Municipality	181,966	Nakuru	
13.	Kitui	Municipality	155,896	Kitui	
14.	Machakos	Municipality	150,041	Machakos	
15.	Mlolongo	Municipality	139,853	Machakos	

Average temperatures for **Nairobi**

	January	February	March	April	May	June	July	August	Sep- tember	October	No- vember	Decem- ber
Avg. Temperature °C	19.7 °C	20.5 °C	20.5 °C	19.3 °C	18.5 °C	17.6 °C	17 °C	17.3 °C	18.5 °C	19.3 °C	18.3 °C	18.6 °C
(°F)	(67.5) °F	(68.9) °F	(68.9) °F	(66.8) °F	(65.3) °F	(63.8) °F	(62.7) °F	(63.1) °F	(65.3) °F	(66.7) °F	(64.9) °F	(65.5) °F
Min. Temperature °C	14.4 °C	14.8 °C	15.5 °C	15.7 °C	14.7 °C	13.3 °C	12.2 °C	12.6 °C	13.1 °C	14.5 °C	14.7 °C	14.4 °C
(°F)	(57.8) °F	(58.7) °F	(59.9) °F	(60.2) °F	(58.5) °F	(56) °F	(54) °F	(54.6) °F	(55.7) °F	(58.1) °F	(58.4) °F	(57.8) °F
Max. Temperature °C	25.8 °C	26.8 °C	26.5 °C	24.4 °C	23.1 °C	22.2 °C	22 °C	22.3 °C	24.3 °C	25 °C	23.2 °C	24 °C
(°F)	(78.5) °F	(80.3) °F	(79.7) °F	(75.9) °F	(73.6) °F	(72) °F	(71.5) °F	(72.2) °F	(75.7) °F	(77.1) °F	(73.8) °F	(75.2) °F
Precipitation / Rainfall	37	34	62	125	102	48	20	28	20	49	94	55
mm (in)	(1.5)	(1.3)	(2.4)	(4.9)	(4)	(1.9)	(0.8)	(1.1)	(0.8)	(1.9)	(3.7)	(2.2)
Humidity(%)	59%	53%	59%	72%	72%	67%	64%	63%	58%	59%	75%	70%
Rainy days (d)	5	4	9	16	13	7	3	4	3	8	14	9

Heterogeneity of Representations

Why merging different data?

No	City/ Town	Status	Population	County	Ref
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Heterogeneity of Representations

Why merging different data?



nairobi average temperatures



Tutti

Immagini

Maps

Notizie

Shopping

Altro

Impostazioni

Strumenti

Circa 1.100.000 risultati (0,53 secondi)

Nairobi, Kenya

Dati meteo medi

Panoramica

Grafici

Mese	Massima / Minima (°C)	Pioggia
Gennaio	26° / 16°	4 giorni
Febbraio	27° / 16°	3 giorni
Marzo	27° / 16°	7 giorni
Aprile	25° / 16°	12 giorni
Maggio	24° / 15°	9 giorni
Giugno	23° / 14°	2 giorni

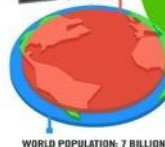
Temperatures for Nairobi

June	July	August	September	October	November	December
17.6 °C (63.8) °F	17 °C (62.7) °F	17.3 °C (63.1) °F	18.5 °C (65.3) °F	19.3 °C (66.7) °F	18.3 °C (64.9) °F	18.6 °C (65.5) °F
13.3 °C (56) °F	12.2 °C (54) °F	12.6 °C (54.6) °F	13.1 °C (55.7) °F	14.5 °C (58.1) °F	14.7 °C (58.4) °F	14.4 °C (57.8) °F
22.2 °C (72) °F	22 °C (71.5) °F	22.3 °C (72.2) °F	24.3 °C (75.7) °F	25 °C (77.1) °F	23.2 °C (73.8) °F	24 °C (75.2) °F
48 (1.9)	20 (0.8)	28 (1.1)	20 (0.8)	49 (1.9)	94 (3.7)	55 (2.2)
67%	64%	63%	58%	59%	75%	70%
7	3	4	3	8	14	9

VV + Veracity

40 ZETTABYTES
[43 TRILLION GIGABYTES]
of data will be created by
2020, an increase of 300
times from 2005

6 BILLION PEOPLE
have cell phones



WORLD POPULATION: 7 BILLION

Volume SCALE OF DATA

It's estimated that
2.5 QUINTILLION BYTES
[2.3 TRILLION GIGABYTES]
of data are created each day



Most companies in the
U.S. have at least
100 TERABYTES
[100,000 GIGABYTES]
of data stored

The New York Stock Exchange
captures
**1 TB OF TRADE
INFORMATION**
during each trading session



By 2016, it is projected
there will be
**18.9 BILLION
NETWORK
CONNECTIONS**
— almost 2.5 connections
per person on earth



Velocity ANALYSIS OF STREAMING DATA

Modern cars have close to
100 SENSORS
that monitor items such as
fuel level and tire pressure



The FOUR V's of Big Data

From traffic patterns and music downloads to web history and medical records, data is recorded, stored, and analyzed to enable the technology and services that the world relies on every day. But what exactly is big data, and how can these massive amounts of data be used?

As a leader in the sector, IBM data scientists break big data into four dimensions: **Volume, Velocity, Variety and Veracity**

Depending on the industry and organization, big data encompasses information from multiple internal and external sources such as transactions, social media, enterprise content, sensors and mobile devices. Companies can leverage data to adapt their products and services to better meet customer needs, optimize operations and infrastructure, and find new sources of revenue.

By 2015
4.4 MILLION IT JOBS
will be created globally to support big data,
with 1.9 million in the United States



As of 2011, the global size of
data in healthcare was
estimated to be
150 EXABYTES
[161 BILLION GIGABYTES]



**30 BILLION
PIECES OF CONTENT**
are shared on Facebook
every month



Variety DIFFERENT FORMS OF DATA

By 2014, it's anticipated
there will be
**420 MILLION
WEARABLE, WIRELESS
HEALTH MONITORS**

**4 BILLION+
HOURS OF VIDEO**
are watched on
YouTube each month



400 MILLION TWEETS
are sent per day by about 200
million monthly active users



**1 IN 3 BUSINESS
LEADERS**
don't trust the information
they use to make decisions



in one survey were unsure of
how much of their data was
inaccurate



Veracity UNCERTAINTY OF DATA

Poor data quality costs the US
economy around
\$3.1 TRILLION A YEAR



Semantic Data Enrichment: from Interactive Exploration to Scalable Deployment

Roberto Avogadro *, Flavio De Paoli ^, Dumitru Roman *, Matteo Palmonari ^

Part II: Semantic Data Enrichment, Applications and Requirements



This work presented in this presentation has received funding from the European Union's Horizon 2020 research and innovation program under grant agreements No 732590 - **EW-Shopp** - and No 732003 – **euBusinessGraph** - and from the European Union's Horizon Europe research and innovation program under grant agreements No 101070284 - **enRichMyData**.



Data Enrichment vs Semantics

- Part of data preparation
 - Goal: add context to an input data set
 - Related to data cleaning (blurred boundaries)
- Structure and terminology
 - Input: a data set D
 - Output: a data set D' with additional data
 - fetched from external data sources
 - inferred / computed
 - ... also “data augmentation” (tables) and “data annotation” (tables and documents)
- The role of semantics
 - Integration-related tasks and annotations
 - Knowledge Graphs (KGs) as enrichment sources

Categories	Available features
Data discovery	Locate missing values (nulls)
	Locate outliers
	Search by pattern
	Sort data
Data validation	Compare values (selection and join)
	Check data range
	Check permitted characters
	Check column uniqueness
	Find type-mismatched data
	Find data-mismatched datatypes
Data structuring	Change column data type
	Delete column
	Detect & change encoding
	Pivot / unpivot
	Rename column
	Split column
	Transform by example [13]
Data enrichment	Assign semantic data type
	Calculate column using expressions
	Discover & merge external data
	Duplicate column
	Generate primary key column
	Join & union
	Merge columns
	Normalize numeric values
Data filtering	Delete/keep filtered rows
	Delete empty and invalid rows
	Extract value parts
	Filter with regular expressions
Data cleaning	Change date & time format
	Change letter case
	Change number format
	Deduplicate data
	Delete by pattern
	Edit & replace cell data
	Fill empty cells
	Remove extra whitespace
	Remove diacritics
	Standardize strings by pattern
	Standardize values in clusters

Examples from the Industry

Domain	Value	Enrichment Data Sources	Data
eCommerce	Predict impact of events on customer searches 	<u>Events</u> , weather	Tabular
Retail	Workforce/budget optimization 	<u>Events</u> , weather	Tabular
CRM	Workforce optimization 	<u>Events</u> , <u>weather</u>	Tabular
IOT	Customer flow analysis 	Events, weather	Tabular
Digital Marketing	Ad impression prediction for campaign optimization 	Weather	Tabular
Digital Marketing	Ad impression prediction for campaign optimization 	Events	Tabular
Manufacturing	AI-based analytics on welding robot data (tables and user manuals)	Proprietary ~KG	Tabular, Texts
Manufacturing	Troubleshooting and repair based on service manuals, records, log data	Proprietary ~KG	Tabular, Texts
Open data	Construction and maintenance of a European dataset of organizations in procurement from tenders	Proprietary ~KG, Wikidata, Crunch Base	Tabular, Texts
Observatory on AI	Construction and maintenance of a KG to track AI-related innovations from different data sources	Crunch Base, WikiData	Tabular, Texts
Business analysis	Cost-effective enrichment of client datasets' with proprietary company KG	Proprietary KG	Tabular

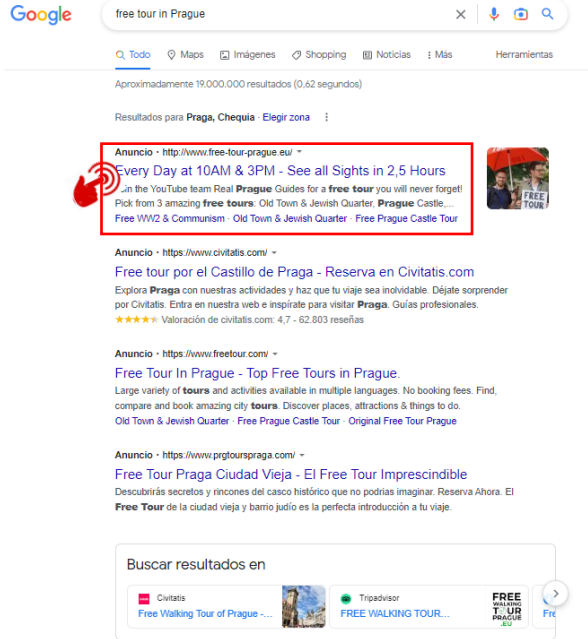
BC1: Marketing data Enrichment for smart-bidding optimization

(MarkEn)

Fernando Perales
Sergio Gonzalo
Elena Morales

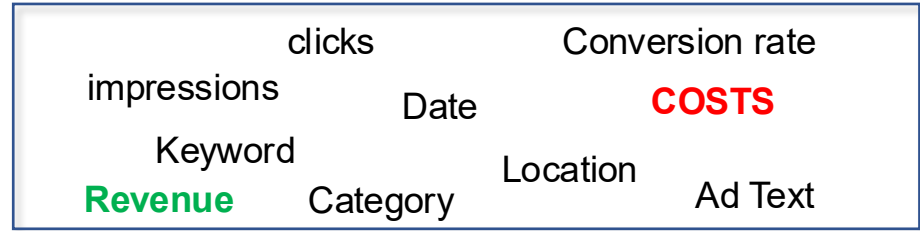


Motivation

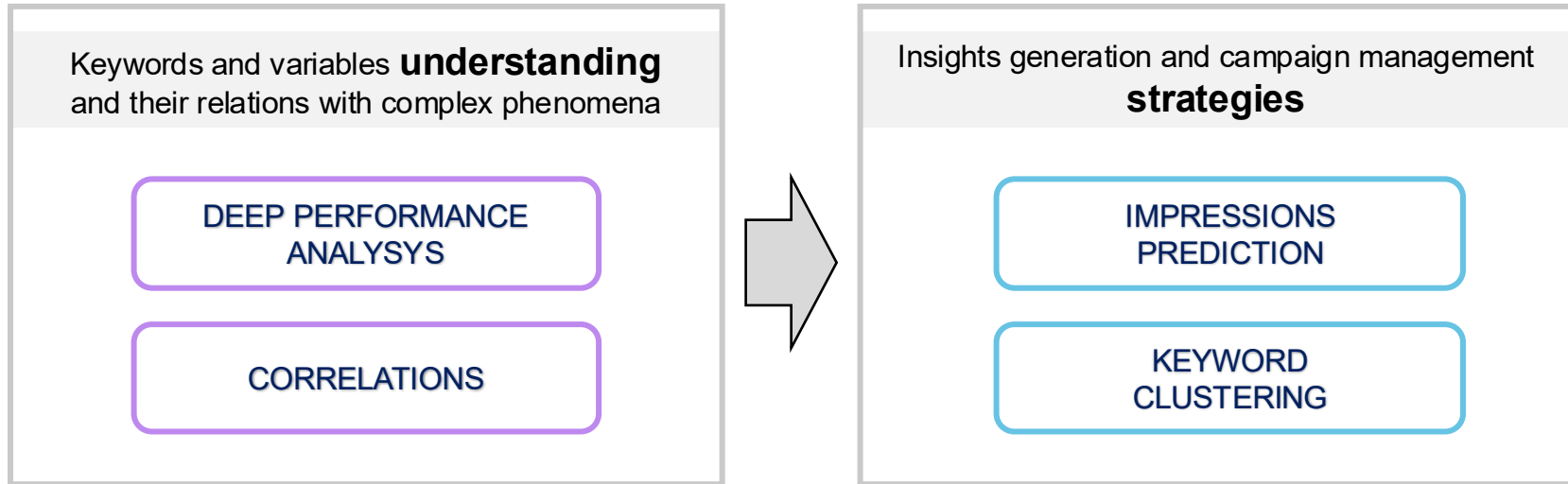


Google search results for "free tour in Prague". The search bar shows "free tour in Prague" with a search icon. Below the search bar, there are tabs for "Todo", "Maps", "Imágenes", "Shopping", "Noticias", "Más", and "Herramientas". The results show approximately 19,000,000 results in 0.62 seconds. The first result is an advertisement for "Every Day at 10AM & 3PM - See all Sights in 2,5 Hours" from "http://www.free-tour-prague.eu/". The second result is an advertisement for "Free tour por el Castillo de Praga - Reserva en Civitatis.com". The third result is an advertisement for "Free Tour In Prague - Top Free Tours in Prague." from "http://www.freetour.com/". The fourth result is an advertisement for "Free Tour Praga Ciudad Vieja - El Free Tour Imprescindible" from "http://www.prgtourspraga.com/". The sidebar on the right shows "Buscar resultados en" with options for "Civitatis", "Tripadvisor", and "FREE WALKING TOUR...".

Performance Data



Downstream Problem



MarkEn Data Sets

Google Cloud

Select a project

Search (/) for resources, docs, products and more

Q Search

Cloud Storage

Bucket details

Go to path

Refresh

Overview

Buckets

Monitoring

Settings

Storage Intelligence

Insights datasets

Cloud Marketplace

Release notes

Folder browser

enrich_my_data

2022/

10_Octubre/

11_Noviembre/

12_Diciembre/

2023/

01_Enero/

02_Febrero/

03_Marzo/

04_Abril/

05_Mayo/

06_Junio/

07_Julio/

08_Agosto/

09_Septiembre/

Buckets

enrich_my_data

2023

01_Enero

Create folder

Upload

Transfer data

Other services

Learn

Filter by name prefix only

Filter

Filter objects and folders

Show

Live objects only

Name	Size	Type	Created	Storage class	Last modified	Public access	Version
EMD_January_001.csv	209.8 MB	text/csv	20 Nov 2024, 21:11:39	Nearline	20 Nov 2024, 21:11:39	Not public	
EMD_January_002.csv	209.4 MB	text/csv	20 Nov 2024, 21:11:39	Nearline	20 Nov 2024, 21:11:39	Not public	
EMD_January_003.csv	209 MB	text/csv	20 Nov 2024, 21:11:39	Nearline	20 Nov 2024, 21:11:39	Not public	
EMD_January_004.csv	209.8 MB	text/csv	20 Nov 2024, 21:11:44	Nearline	20 Nov 2024, 21:11:44	Not public	
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EMD_January_006.csv	209.6 MB	text/csv	20 Nov 2024, 21:11:46	Nearline	20 Nov 2024, 21:11:46	Not public	
EMD_January_007.csv	209.6 MB	text/csv	20 Nov 2024, 21:11:50	Nearline	20 Nov 2024, 21:11:50	Not public	
EMD_January_008.csv	209.6 MB	text/csv	20 Nov 2024, 21:11:50	Nearline	20 Nov 2024, 21:11:50	Not public	
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EMD_January_010.csv	210.3 MB	text/csv	20 Nov 2024, 21:11:55	Nearline	20 Nov 2024, 21:11:55	Not public	
EMD_January_011.csv	209.9 MB	text/csv	20 Nov 2024, 21:11:55	Nearline	20 Nov 2024, 21:11:55	Not public	
EMD_January_012.csv	210 MB	text/csv	20 Nov 2024, 21:11:56	Nearline	20 Nov 2024, 21:11:56	Not public	
EMD_January_013.csv	211 MB	text/csv	20 Nov 2024, 21:12:00	Nearline	20 Nov 2024, 21:12:00	Not public	
EMD_January_014.csv	210.3 MB	text/csv	20 Nov 2024, 21:12:00	Nearline	20 Nov 2024, 21:12:00	Not public	
EMD_January_015.csv	210.6 MB	text/csv	20 Nov 2024, 21:12:02	Nearline	20 Nov 2024, 21:12:02	Not public	
EMD_January_016.csv	210.3 MB	text/csv	20 Nov 2024, 21:12:06	Nearline	20 Nov 2024, 21:12:06	Not public	
EMD_January_017.csv	210.7 MB	text/csv	20 Nov 2024, 21:12:06	Nearline	20 Nov 2024, 21:12:06	Not public	
EMD_January_018.csv	211 MB	text/csv	20 Nov 2024, 21:12:08	Nearline	20 Nov 2024, 21:12:08	Not public	
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EMD_January_021.csv	210.3 MB	text/csv	20 Nov 2024, 21:12:13	Nearline	20 Nov 2024, 21:12:13	Not public	
EMD_January_022.csv	211.5 MB	text/csv	20 Nov 2024, 21:12:17	Nearline	20 Nov 2024, 21:12:17	Not public	
EMD_January_023.csv	211.3 MB	text/csv	20 Nov 2024, 21:12:17	Nearline	20 Nov 2024, 21:12:17	Not public	
EMD_January_024.csv	210.7 MB	text/csv	20 Nov 2024, 21:12:19	Nearline	20 Nov 2024, 21:12:19	Not public	

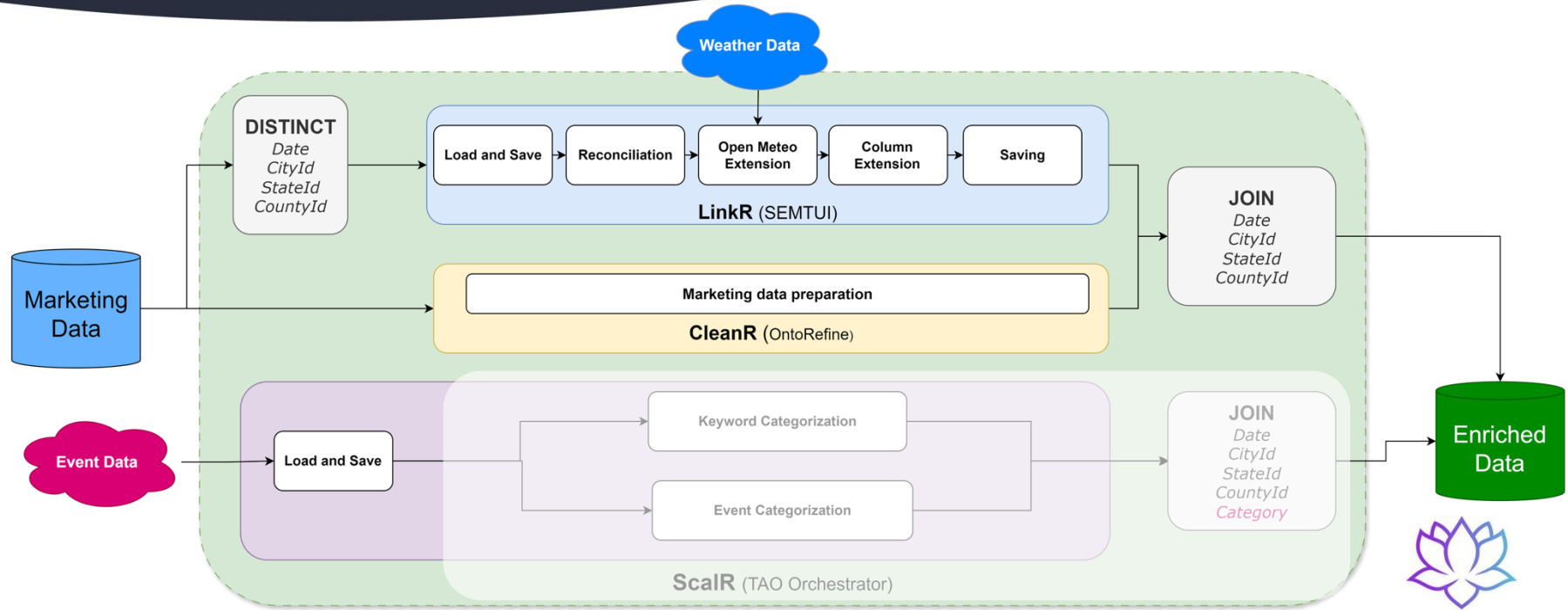
35 files per month
27 moths period
200 Mb per file



189 Gb of data to
be processed



JOT's Data Preparation Workflow



Making Sense of Digital Marketing Analytics: External Events



Figure 12: Correlation between X and online digital indicators in the "SportFitness" category.

Making Sense of Digital Marketing Analytics: External Events

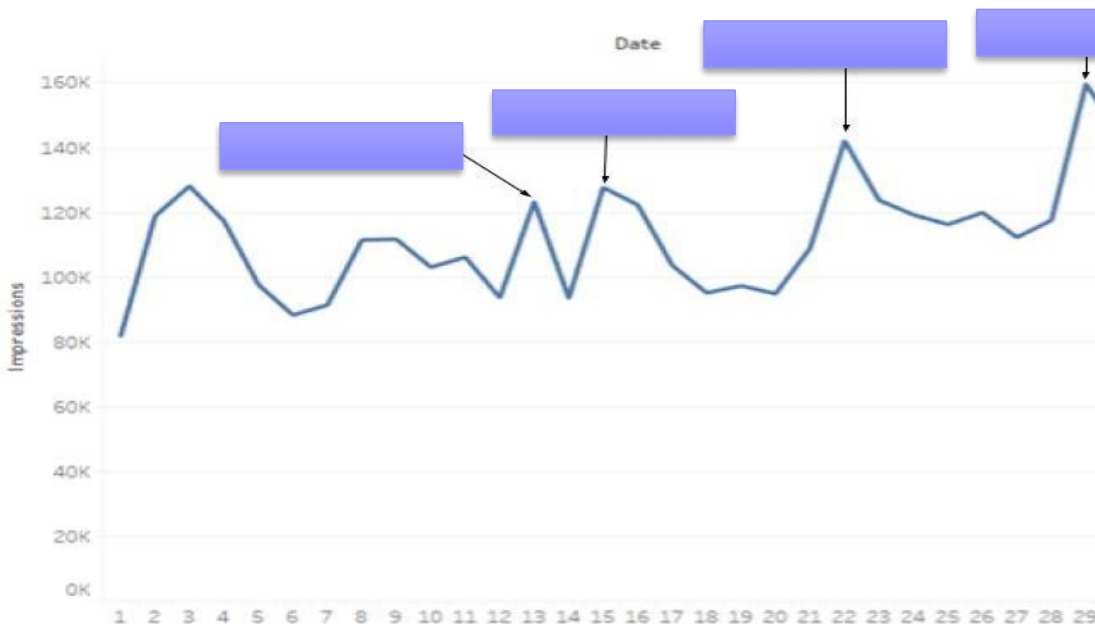


Figure 12: Correlation between X and indicators in the "SportFitness" category.

germany january 2017

Tutti Immagini Notizie Video Maps Altro Impostazioni Strumenti

Circa 57.000.000 risultati (0,58 secondi)

Upcoming Events in Germany (January 2018), Events Happening in ...
<https://10times.com/germany?month=january> Traduci questa pagina
 Events in Germany in January. Follow 14,945 ... Januar 2018 at the ESTREL Convention Center in Berlin, Germany. This event ... PSI 2017 (PSI Dusseldorf).

Immagini relative a germany january 2017

→ Altre immagini per germany january 2017 Segnala immagini non appropriate

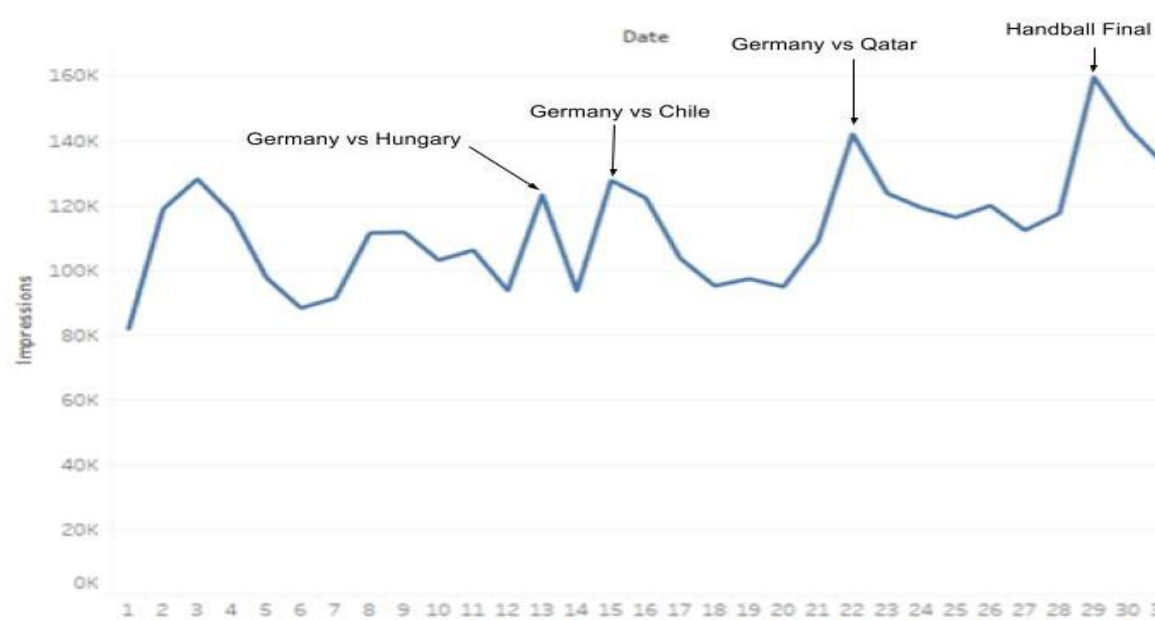
A Month of Islam and Multiculturalism in Germany: January 2017
<https://www.gatestoneinstitute.org/9959/germany-islam-january> Traduci questa pagina
 19 feb 2017 - "If we are serious about the fight against Islamism and terrorism, then it must also be a cultural struggle." — German Vice Chancellor Sigmar ...

2017 in Germany - Wikipedia
https://en.wikipedia.org/wiki/2017_in_Germany Traduci questa pagina
 Passa a **January** - January[edit]. January 10: Roman Herzog, German politician, former president (born 1934); Januar 13: Udo Ulfkotte, German journalist ...
 Events Deaths

Berlin January Weather 2017 - AccuWeather Forecast for Berlin ...
<https://www.accuweather.com/en/de/berlin/.../january.../178087> Traduci questa pagina
 Get Berlin, Germany typical January Weather including average and record temperatures from AccuWeather.com.

Public Holidays in Germany in 2017 | Office Holidays
www.officeholidays.com/countries/germany/index.php Traduci questa pagina
 List of National Public Holidays celebrated in Germany during 2017 with information on the ... 3 October 2017 This National holiday is in 14 days. ... Friday, January 06 Jan 6, Three King's Day, Baden-Württemberg, Bavaria, Saxony-Anhalt.

Making Sense of Digital Marketing Analytics: External Events



WORLD MEN'S HANDBALL CHAMPIONSHIP MUST SEE MATCHES

By BeIN SPORTS January 11, 2017 11:38 AM

As the 2017 World Handball Championship in France draws near, Sports fans in general and handball fans in particular are looking forward to intense first round showdowns that rival knockout stages.

Figure 12: Correlation between German national handball team matches and online digital indicators in the "SportFitness" category.

Spend Network:

**Easy access and analysis for
the world's \$13trn government
contracting market.**

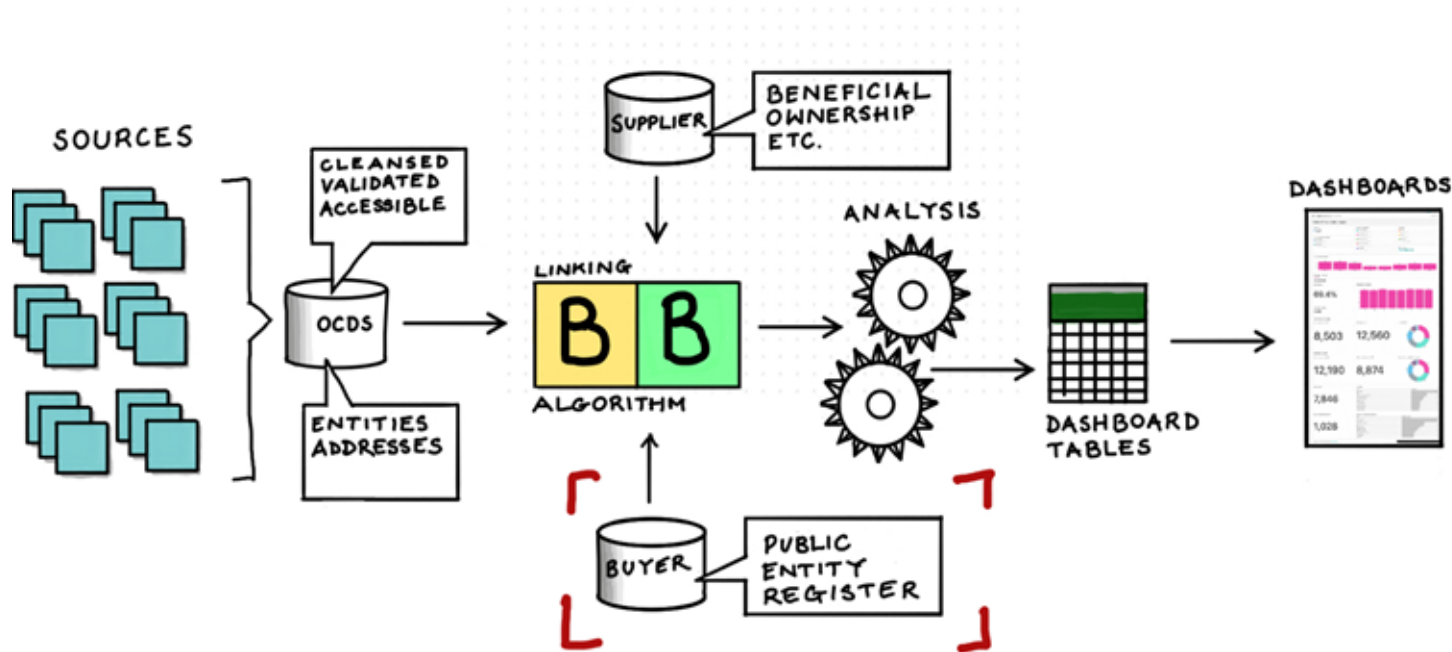
Context

1. Our challenge & Procurement data
 - a. Context / quality is poor
2. Goal -> Better product -> Enrichment
3. Production
4. Results

Existing enrichments

1. Categories (CPV - Level 3)
2. Language (ISO - 639)
3. Continent & Country (ISO 3166)
4. Currency (ISO 4217)

Existing ETL & Enrichments



Integration & Interoperability

Mona Lisa (Q12418)

Item Discussion

oil painting by Leonardo da Vinci in the Louvre, Paris
La Gioconda | Monna Lisa | The Mona Lisa

► In more languages

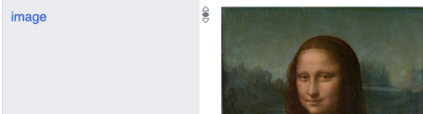
Large-scale encyclopedic KGs

Statements

<https://www.wikidata.org/wiki/Q12418>

instance of painting
► 1 reference

inception 1500s
earliest date 1503
latest date 1506
► 1 reference



significant event

vandalism
point in time 1956
cause of destruction acid attack
► 1 reference

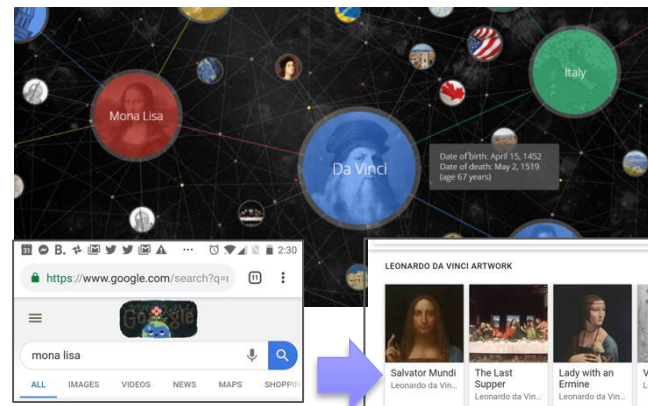
vandalism
point in time 30 December 1956
cause of destruction stone throwing
► 1 reference

vandalism
point in time 2 April 2009
cause of destruction throwing
► 0 references

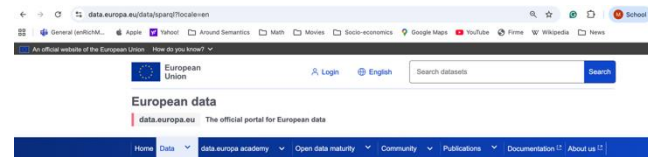
acquisition
point in time 1519 Gregorian
beforehand owned by Leonardo da Vinci
afterward owned by Francis I of France
► 1 reference

theft
point in time 21 August 1911 Gregorian
perpetrator Vincenzo Peruggia
location Louvre Palace
end time 10 December 1913 Gregorian
statement is subject of theft of Mona Lisa
► 1 reference

vandalism
point in time 29 May 2022
cause of destruction pieing
► 1 reference



One source for many applications



SPARQL search

All metadata of data.europa.eu is stored as RDF triples (RDF) and can be queried using SPARQL query language at this endpoint. Queries can be executed via the query editor (below) or via the machine-readable endpoint (<https://data.europa.eu/sparql>).

To assist users in constructing useful SPARQL queries we provide a page with featured sample SPARQL queries that can be executed via our endpoint. Datasets and ontologies published as RDF triples on the portal follow the DCAT-AP specification. For further reading on the metadata ontology, please refer to the data.europa.eu academy page on DCAT-AP at: <https://data.europa.eu/en/academy/dcat-end-dcat-ap>.

```
Query =
1 PREFIX dcat: <http://www.w3.org/ns/dcat#>
2 PREFIX org: <http://data.europa.eu/ontology/ontology/ontology#>
3 PREFIX dci: <http://purl.org/dc/terms/>
4 PREFIX xsl: <http://www.w3.org/2001/XMLSchema#>
5 PREFIX foaf: <http://xmlns.com/foaf/0.1/>
```

<https://data.europa.eu/data/sparql?locale=en>

Open data portals

36

- yago-res:Mona Lisa
- <http://sw.cyc.com/concept/Mx4rv573qZwpEbGdrcN5Y29ycA>
- freebase:Mona Lisa
- dbpedia-de:Mona Lisa
- <http://d-nb.info/gnd/4074156-4>
- wikidata:Mona Lisa

Integration & Interoperability

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► In more languages

Large-scale encyclopedic KGs

Statements

<https://www.wikidata.org/wiki/Q12418>

instance of painting

► 1 reference

inception 1500s

earliest date 1503
latest date 1506

► 1 reference

image



Links to
same entity
across
different
sources

- [yago-res:Mona Lisa](#)
- <http://sw.cyc.com/concept/Mx4rv573qZwpEbGdrcN5Y29ycA>
- [freebase:Mona Lisa](#)
- [dbpedia-de:Mona Lisa](#)
- <http://d-nb.info/gnd/4074156-4>
- [wikidata:Mona Lisa](#)

https://dbpedia.org/page/Mona_Lisa

significant event

vandalism
point in time 1956
cause of destruction

► 1 reference

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► 0 references

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afterward owned by Leonardo da Vinci
Francis I of France

► 1 reference

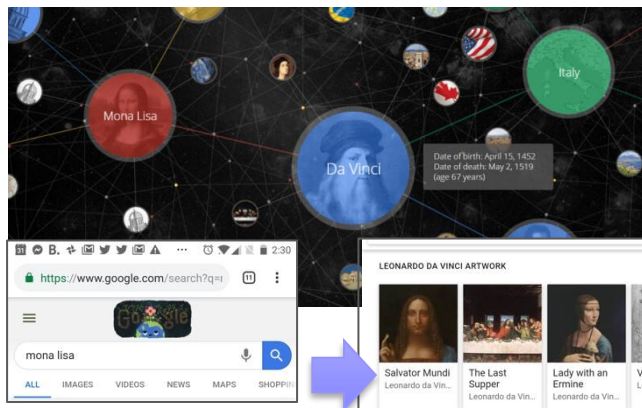
theft
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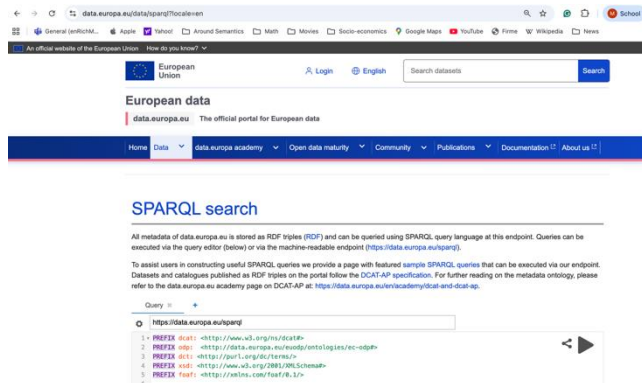
vandalism
point in time 29 May 2022
cause of destruction pieing

► 1 reference

Modeling
using shared
vocabularies



One source for many applications



<https://data.europa.eu/data/sparql?locale=en>

Open data portals

Data Semantics / Why

■ When should we take care of data semantics?

- When we need to **integrate** different and heterogenous data sources
- When we need to **make a dataset accessible** to parties who do not know the dataset
- When we need to **access** a dataset that is produced by third parties
- When we need to **infer meaning of data**

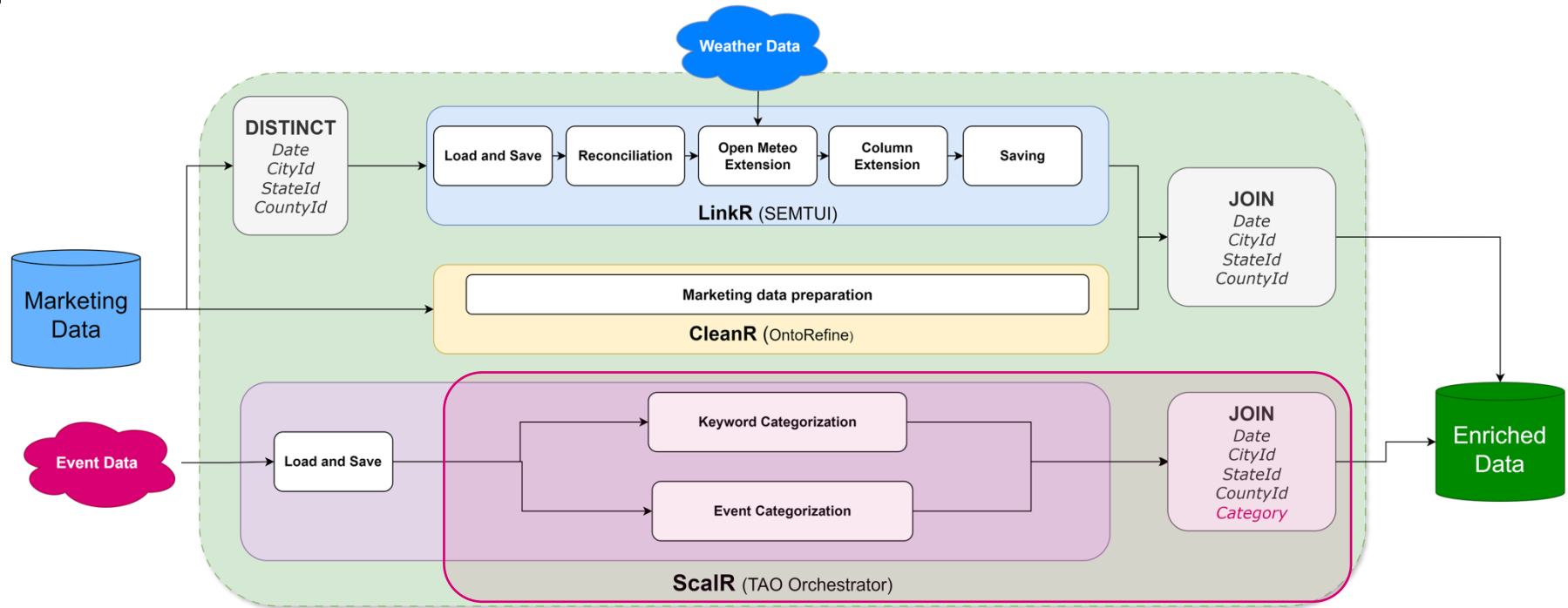
■ Why?

- To **improve data processing** in crucial steps of the data science lifecycle
- **Data preparation** (including: cleaning, integration, transformation) can cover up to 80% of effort in a data science project

■ Data Semantics vs. AI

- Several overlapping topics, more focus on data for AI

JOT's Data Preparation Workflow



Large Language Models (LLMs)

- Essentially: probability distribution estimators
 - complex approaches based on Neural Networks to determine the probability of a given word in a given context
 - **massive amount of parameters** learned after «reading» **massive amounts of text**
 - context input affect the probability of what comes next/before
- Plus some more tricks
 - Picking words different from the most probable one (for language generation)
 - Instruct to pick up answers that maximize certain properties (InstructGPT, ChatGPT)
 - Reasoning (CoT, space exploration,...)

Size examples

From the Meta's LLAMA language models -

<https://ai.facebook.com/blog/large-language-model-llama-meta-ai/> -

trained only on open data

Data

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

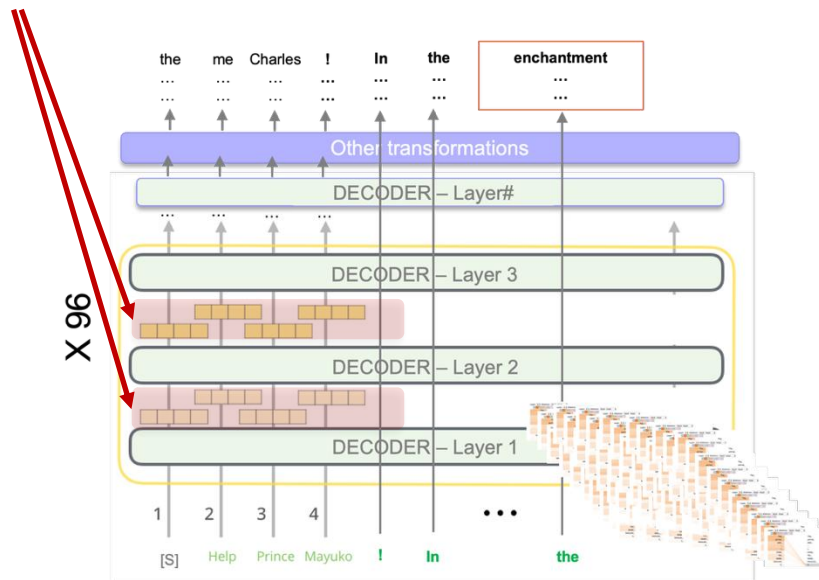
Parameters

- 6.7B, 13.0B, 32.5B, 65.2B

Large Language Models (LLMs)

- Essentially: probability distribution estimators
 - complex approaches based on Neural Networks to determine the probability of a given word in a given context
 - **massive amount of parameters** learned after «reading» **massive amounts of text**
 - context input affect the probability of what comes next/before
- Plus some more tricks
 - Picking words different from the most probable one (for language generation)
 - Align answers to human preferences and Instruct to pick up answers that maximize certain properties (InstructGPT, ChatGPT)
 - Reasoning (CoT, space exploration,...)

Infer meaning of data



LLM based on Transformer (decoder)

Data Semantics / What (the intuition)

■ What is data semantics?

□ Data semantics is about

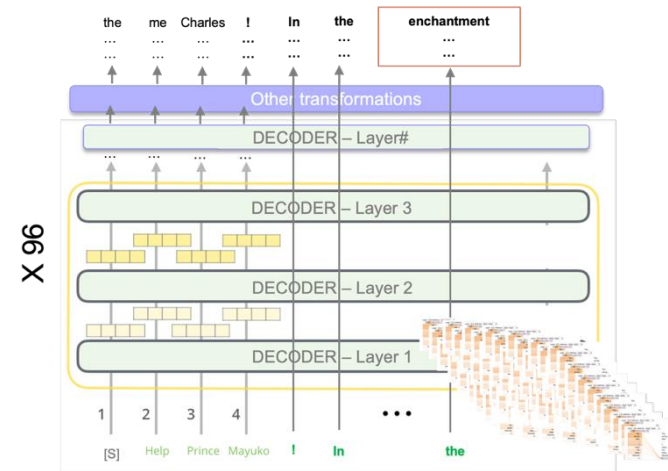
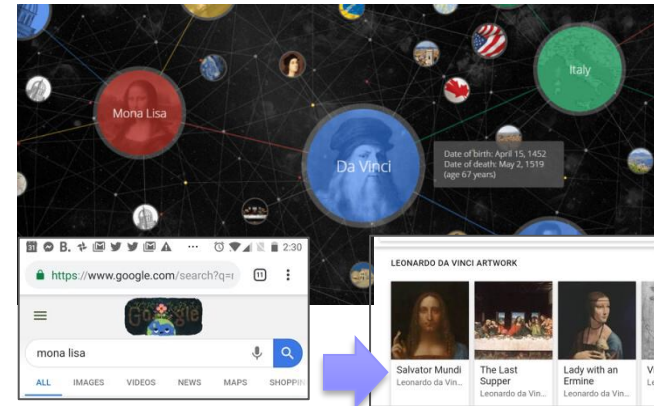
- **modeling the meaning** of data and using it to improve their interpretation, interoperability across systems, and subsequent processing

- How can we model data better so that ...
- Symbolic AI a.k.a. Declarative AI, **Knowledge Representation and Reasoning (KRR)**, Knowledge Engineering

- **interpreting the meaning** of data and use this imputed interpretation to improve data processing

- How can we interpret data better so that ...
- Neural AI a.k.a. Deep Learning, **Representation Learning**, Language Models, Generative AI

Two different semantic paradigms



Data Semantics / What (the intuition)

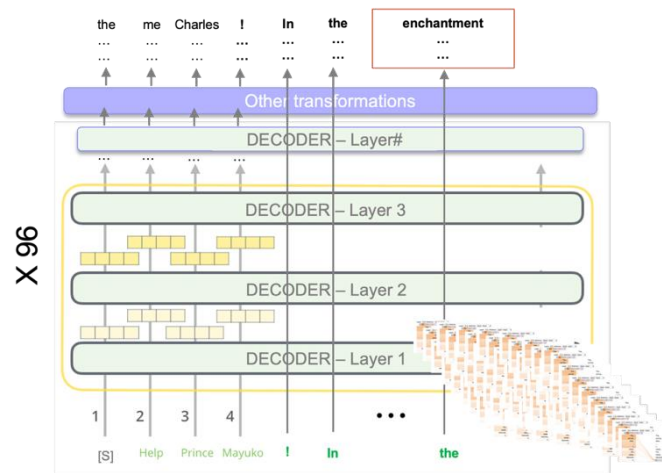
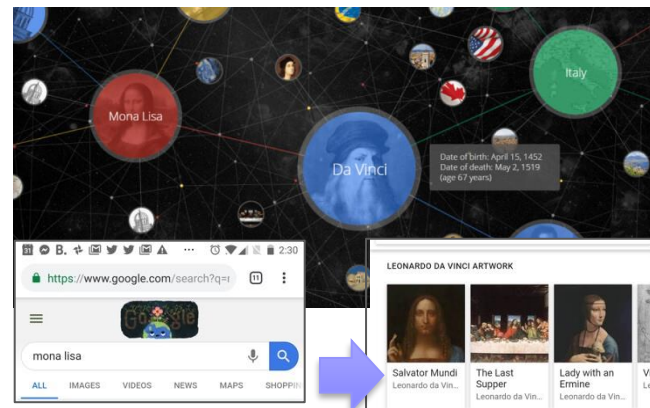
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Do we still need to care about data modeling, in the era of LLMs?



Advances: NLP and Large Language Models

2011: IBM's Watson beats Jeopardy's champions

Question Answering



2019: 100M of Alexa have been sold

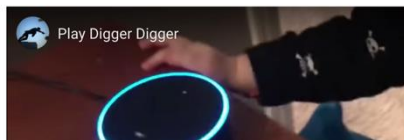


Digger digger kid

In most cases, virtual assistants pick up what you're saying fairly accurately. However, some parents have found out that that isn't always the case.

One unknown YouTube user uploaded footage of his son asking their Alexa device to "play Digger, Digger". Mishearing the toddler, Alexa's delivers a far from PG response as his parents scream for her to stop.

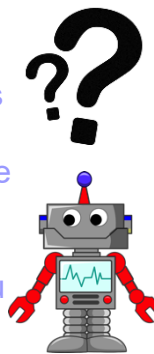
We're warning you - it's probably best not to watch this one at work:



Italo Calvino's Invisible Cities:
which one is AI-generated?

"Arrivando a ogni nuova città il viaggiatore ritrova un suo passato che non sapeva più d'avere: l'estraneità di ciò che non sei più o non possiedi più t'aspetta al varco nei luoghi estranei e non posseduti."

"By reaching every new city, the traveler finds a past that no longer knew: The disenchantment of what you are no longer, or you no longer have, awaits the crossing in foreign and non-owned places."



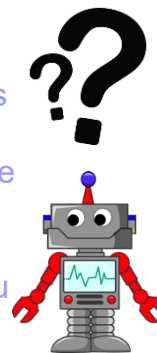
"Arriving at each new city, the traveler finds again a past of his that he did not know he had: the foreignness of what you no longer are or no longer possess lies in wait for you in foreign, unpossessed places."

Advances: NLP and Large Language Models

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“Arriving at each new city, the traveler finds again a past of his that he did not know he had: the foreignness of what you no longer are or no longer possess lies in wait for you in foreign, unpossessed places.”

ChatGPT 2025

“Arriving at each new city, the traveler finds again a part of his past that he did not know he had: the strangeness of what you no longer are or no longer possess awaits you at the threshold in unfamiliar places, in places not your own.””

Advances: NLP and Large Language Models

Do you know the source of this piece of text?

"Arrivando a ogni nuova città il viaggiatore ritrova un suo passato che non sapeva più d'avere: l'estraneità di ciò che non sei più o non possiedi più t'aspetta al varco nei luoghi estranei e non posseduti"

Yes! This passage is from *Le città invisibili* (*Invisible Cities*) by **Italo Calvino**. The book, published in 1972, is a poetic and philosophical novel structured as a fictional dialogue between Marco Polo and Kublai Khan, where Polo describes imaginary cities with deep reflections on memory, desire, and perception.

This specific excerpt reflects on how travelers rediscover parts of their past in new cities, emphasizing the relationship between memory and places.

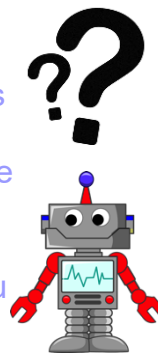
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Italo Calvino's *Invisible Cities*:
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"Arriving at each new city, the traveler finds again a past of his that he did not know he had: the foreignness of what you no longer are or no longer possess lies in wait for you in foreign, unpossessed places."

ChatGPT & Data Semantics

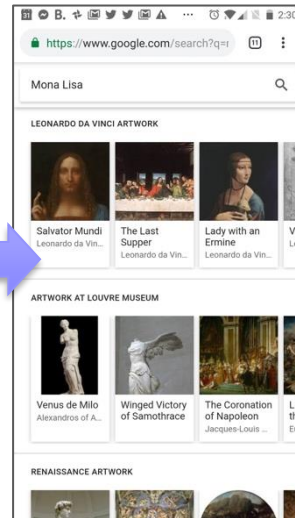
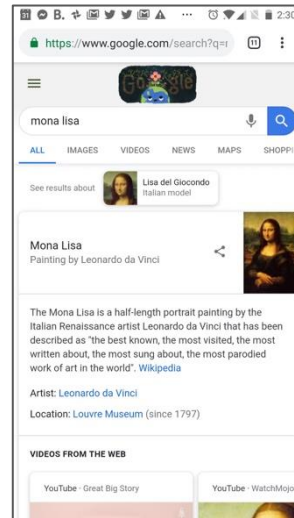
Tell me about mona lisa

The *Mona Lisa* is one of the most famous paintings in the world, created by the Italian artist **Leonardo da Vinci** between **1503 and 1506** (though some believe he worked on it until **1517**). It is a portrait of a woman with a mysterious smile, and it is housed in the **Louvre Museum** in Paris.

Key Facts About the Mona Lisa:

- **Subject:** The painting is believed to depict **Lisa Gherardini**, the wife of Florentine merchant **Francesco del Giocondo**, which is why it's also called *La Gioconda* (in Italian) or *La Joconde* (in French).
- **Medium:** Oil on a **poplar wood panel**.
- **Size:** Relatively small, about **77 cm × 53 cm (30 in × 21 in)**.
- **Technique:** Da Vinci used **sfumato**, a technique that creates soft transitions between colors and tones, adding depth and realism.
- **Expression:** The subject's enigmatic smile and gaze that seems to follow the viewer have intrigued people for centuries.
- **Theft and Fame:** It became world-famous after being **stolen in 1911** by an Italian named **Vincenzo Peruggia**, who believed it should be returned to Italy. It was recovered two years later.
- **Damage & Protection:** The painting is behind **bulletproof glass** to protect it from vandalism.

Would you like to know more about its history, mysteries, or artistic techniques? 😊



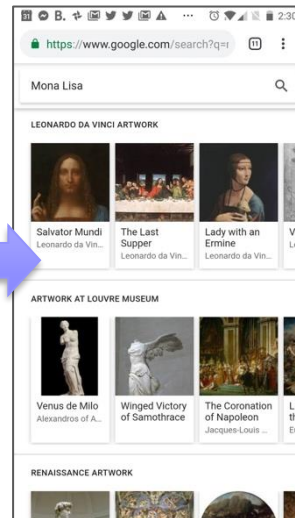
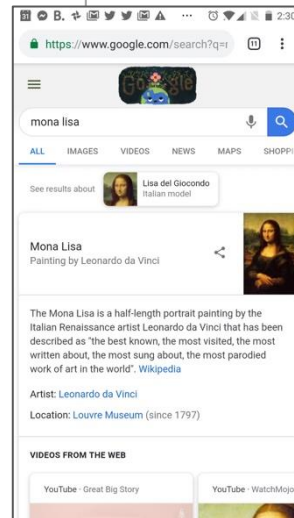
ChatGPT & Data Semantics

Tell me about mona lisa

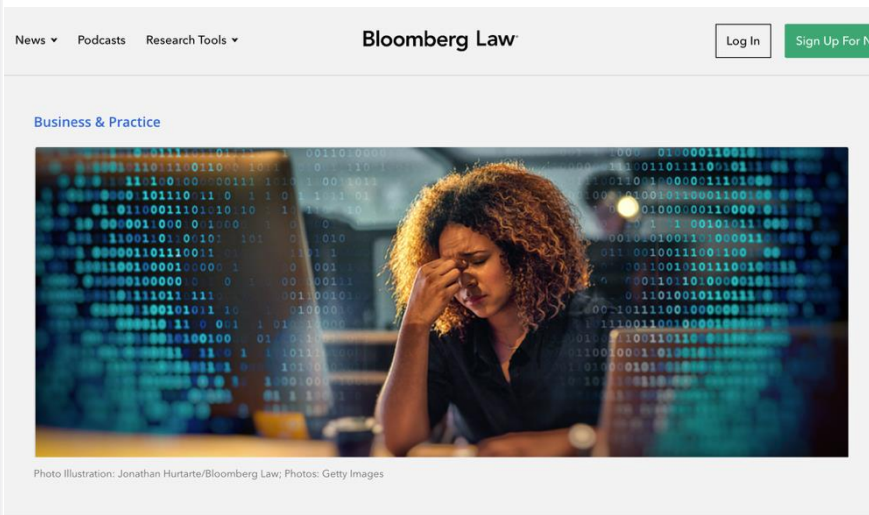
Tell me about other famous paintings hosted in the same museum that hosts Mona Lisa

The *Mona Lisa* is housed in the **Louvre Museum** in Paris, one of the most famous and visited museums in the world. The Louvre contains a vast collection of masterpieces from various periods and cultures. Some other famous paintings in the Louvre include:

- Key**
- 1. The Wedding at Cana – Paolo Veronese**
 - This massive painting (bigger than the *Mona Lisa*) depicts the biblical story of Jesus turning water into wine at a wedding feast.
 - 2. Liberty Leading the People – Eugène Delacroix**
 - A powerful representation of the July Revolution of 1830 in France, with the allegorical figure of Liberty leading the revolutionaries.
 - 3. The Raft of the Medusa – Théodore Géricault**
 - A dramatic and emotional painting depicting the aftermath of a French naval disaster in 1816, where survivors are seen clinging to a makeshift raft.
 - 4. The Coronation of Napoleon – Jacques-Louis David**
 - A grand neoclassical painting illustrating the crowning of Napoleon Bonaparte inside Notre-Dame Cathedral in 1804.
 - 5. The Oath of the Horatii – Jacques-Louis David**
 - A famous neoclassical work showing three Roman brothers swearing allegiance to their father before battle.
 - 6. The Lacemaker – Johannes Vermeer**



Advances: ChatGPT “Passing the Bar Exam”



Feb. 13, 2023, 10:00 AM

ChatGPT Almost Passed the Bar, But Competent Lawyers Do Much More

OPINION



Mary Lu Bilek



Deborah Merritt

Ohio State University, Moritz College of Law

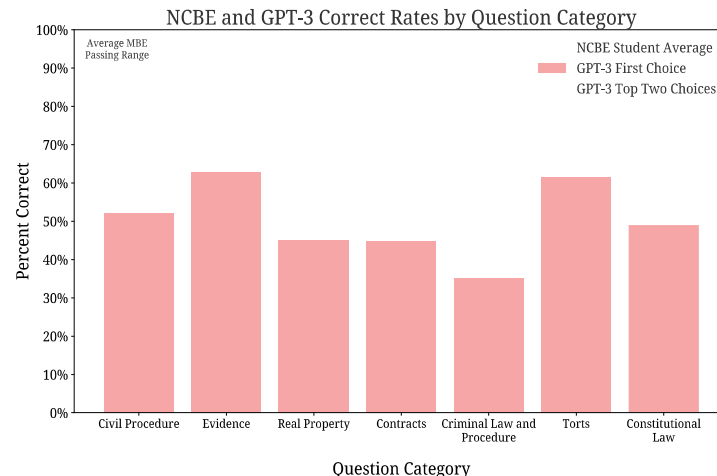


Fig. 1. Summary of performance by question category for GPT -3.5 and NCBE-Reported Students

LLM Failure Reports

ErnestDavis / LLM-Failures Public

<> Code Issues Pull requests Actions Projects Wiki Security Insights

Home

ErnestDavis edited this page on Jan 5 - 9 revisions

Welcome to the LLM-Failures wiki!

This is a collection of failures of recent large language models, particularly ChatGPT and GPT3. We very much welcome contributors. For editing permission, please email Ernest Davis davise@cs.nyu.edu

The Wiki is organized by type of error involved. Each example should be placed on a separate page and linked from the page for the appropriate error type. Please follow the template (or imitate the existing pages).

- [Bias](#)
- [Ethics](#)
- [Logic](#)
- [Math](#)
- [Physical reasoning](#)
- [Psychological reasoning](#)
- [Temporal reasoning](#)
- [Other](#)

<https://github.com/ErnestDavis/LLM-Failures/wiki>

 **giuven95** Add files via upload 519120 3 weeks ago 149 commits

Images	Add files via upload	3 weeks ago
scripts	Update README.md	2 months ago
README.md	Update README.md	3 weeks ago

 README.md

LLM failure archive (ChatGPT and beyond)

A repo containing failure cases for ChatGPT and similar models, with the purpose of further study. I think it could be useful for:

- comparison with other models
- creation of synthetic data for testing and/or training

Click on the underlined sentences to show details

<https://github.com/giujen95/chatgpt-failures>

The Road to AI We Can Trust

Home Archive About



Is it time to hit the pause button on AI?

An essay on technology and policy, co-authored with Canadian Parliament Member Michelle Rempel Garner.

GARY MARCUS AND MICHELLE REMPEL GARNER

FEB 26 54 50

New Top Community



Recommendations

<https://garymarcus.substack.com/>

A Categorical Archive of ChatGPT Failures

Ali Borji
Quintic AI
aliborji@gmail.com

February 24, 2023

<https://arxiv.org/pdf/2302.03494.pdf>

Risks: Factuality & Trust

Here's What Happens When Your Lawyer Uses ChatGPT

A lawyer representing a man who sued an airline relied on artificial intelligence to help prepare a court filing. It did not go well.

<https://www.nytimes.com/2023/05/27/nyregion/avianca-airline-lawsuit-chatgpt.html>

Give this article



1.1K



When Avianca asked a Manhattan federal judge to toss out the case, Mr. Mata's lawyers vehemently objected, submitting a 10-page brief that **cited more than half a dozen relevant court decisions.** There was *Martinez v. Delta Air Lines*, *Zicherman v. Korean Air Lines* and, of course, *Varghese v. China Southern Airlines*, with its learned discussion of federal law and "the tolling effect of the automatic stay on a statute of limitations."

There was just one hitch: No one — not the airline's lawyers, not even the judge himself — could find the decisions or the quotations cited and summarized in the brief.

That was because **ChatGPT had invented everything.**

Risks: Hallucinations

GOOGLE'S BARD

The earliest mention of artificial intelligence (AI) in The New York Times was in 1956, when the term was coined by John McCarthy ¹ at a workshop at Dartmouth College. The workshop brought together some of the brightest minds in computer science to discuss the possibility of creating machines that could think like humans.

The Times reported on the workshop in a front-page article, ² calling it “a new field of science” ³ that was “aimed at creating machines that can think like humans.” The article quoted McCarthy as saying ⁴ that the goal of AI was to “build machines with minds, capable of learning and adapting to new situations.”

NEW YORK TIMES ANALYSIS

- ¹ Mr. McCarthy is credited with creating the term artificial intelligence in 1955, in a proposal for the Dartmouth conference.
- ² The Times did not cover the Dartmouth workshop at all, let alone on the front page.
- ³ More recent online articles refer to the conference as starting “a new field of science.”
- ⁴ The Times could not find a reference to Mr. McCarthy ever saying this phrase, though it includes common ways to describe A.I.

<https://www.nytimes.com/2023/05/01/business/ai-chatbots-hallucination.html>

MICROSOFT'S BING

Based on web search results, the earliest mention of artificial intelligence in the New York Times seems to be from a book review published on March 5, 1956. ¹ The book is called “The Human Use of Human Beings” ² by Norbert Wiener, and the reviewer is John R. Pierce. ³ The review mentions artificial intelligence as one of the topics covered by Wiener, who is described as “a pioneer in cybernetics”.: <https://www.nytimes.com/1956/03/05/archives/the-human-use-of-human-beings-cybernetics-and-society-by-norbert.html> ⁴

NEW YORK TIMES ANALYSIS

- ¹ The Times published a review of a book by Mr. Wiener one day earlier, on March 4, 1956.
- ² This seminal book was published in 1950, not 1956. The March 4 review was of Mr. Wiener's book “I Am A Mathematician: The Later Life of a Prodigy.”
- ³ The March 4 review was not written by John R. Pierce, a noted electrical engineer. It was written by John G. Kemeny, a mathematician.
- ⁴ The format of this URL is consistent with Times archives for articles in this period, but this exact webpage never existed.

BBC vs Gen AIs

Representation of BBC News content in AI Assistants

Foreword

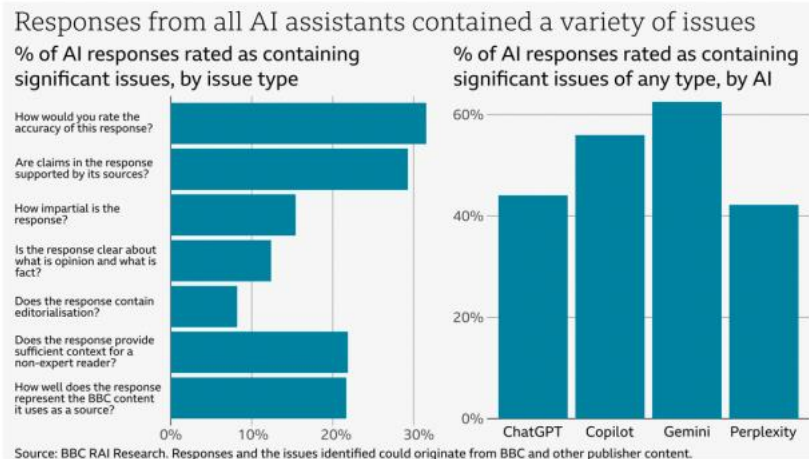
AI assistants risk misleading audiences by distorting BBC Journalism

The answers produced by the AI assistants contained significant inaccuracies and distorted content from the BBC. In particular:

- 51% of all AI answers to questions about the news were judged to have significant issues of some form.
- 19% of AI answers which cited BBC content introduced factual errors – incorrect facts, statements, numbers and dates.
- 13% of the quotes sourced from BBC articles were either altered from the original source or not present in the article cited.

High level findings

51% of all AI responses were flagged as having 'significant issues' of some form by reviewers. 91% of responses contained at least 'some issues'. Journalists analysed responses to all the questions AI attempted to answer, including where no BBC sources were used.



<https://www.bbc.co.uk/aboutthebbc/documents/bbc-research-into-ai-assistants.pdf>

Data Semantics / What (the intuition)

■ What is data semantics?

□ Data semantics is about

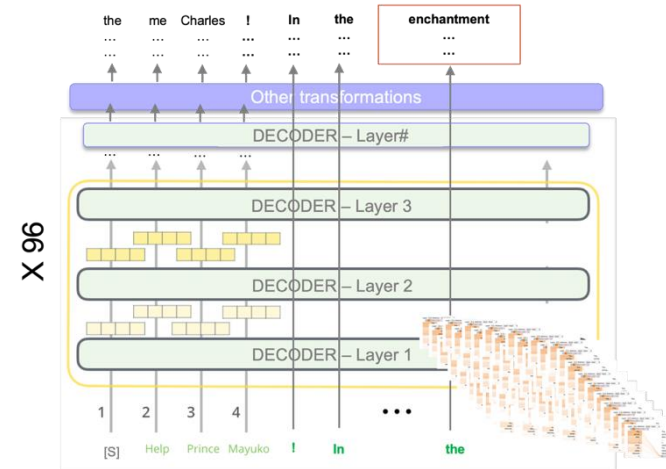
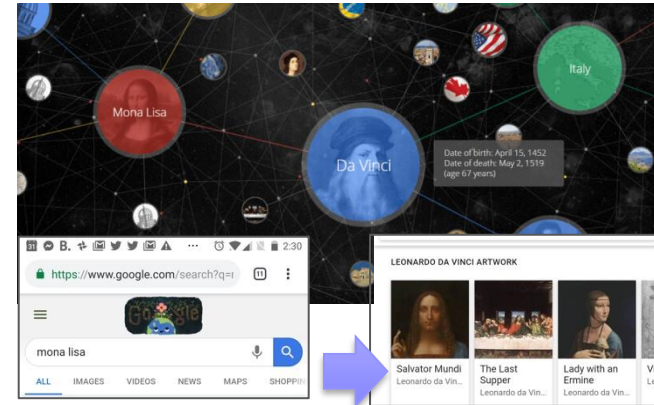
- **modeling the meaning** of data and using it to improve their interpretation, interoperability across systems, and subsequent processing

- How can we model data better so that ...
- Symbolic AI a.k.a. Declarative AI, **Knowledge Representation and Reasoning (KRR)**, Knowledge Engineering

- **interpreting the meaning** of data and use this imputed interpretation to improve data processing

- How can we interpret data better so that ...
- Neural AI a.k.a. Deep Learning, **Representation Learning**, Language Models, Generative AI

Two different semantic paradigms with **strong synergies** in today applications



KRR vs. Neural Networks (incl. LLMs)

■ Semantics in KRR

- Combine truth values according to a (truth) theory
- Only explicit “features” (a model)
- A priori guarantees
 - based on the theory

□ Highlights

- Explicit knowledge
- Brittle to noise/variations
- Precise inferences from input
- + predictability, traceability, expl.
- + control/manipulation

knowledge

+ reliable

■ Semantics in NNs / LLMs

- Learn to combine signals/features from data
- Generate features
- A posteriori guarantees
 - based on empirical assessment

□ Highlights

- Implicit knowledge
- Tolerant to noise/variations
- Best guess from input
- - predictability, traceability, expl.
- - control/manipulation

+ flexible

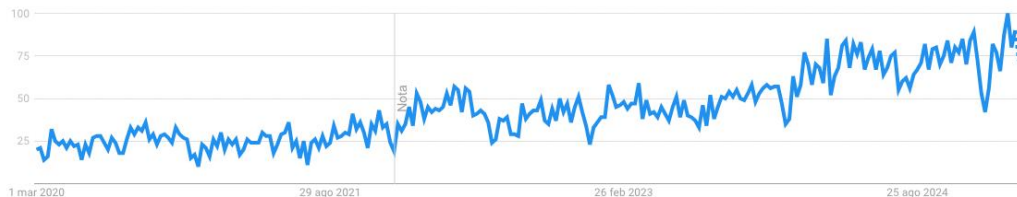
External Knowledge Sources for GenAI

- Very often knowledge requires superhuman reliability
 - Persistence
 - Guarantees on operations
 - Equal treatment of information independently from “frequency”
 - Deterministic inferences
- In practice... databases and knowledge bases
- Then we can use LLMs to interact with these sources in a large variety of ways
 - E.g., context engineering for LLMs with access to sources

Trusted Data and LLMs

- LLMs can help build better data, e.g., graph data extraction
 - We'll see several examples / techniques
- Simple application of LLMs to unstructured data is often not sufficient
- LLMs can work better with “smarter data”

Google Trends for “Knowledge Graphs” from March 2020



Associated queries

In aumento ▼ ⬇ ⬅ ➡ ➦

- | | | |
|---|-----------|-------------|
| 1 | llm | Impennata ⋮ |
| 2 | rag | Impennata ⋮ |
| 3 | graph rag | Impennata ⋮ |

LLMs working with background knowledge graphs or related annotations

Trusted Data and LLMs

- LLMs can help build better data, e.g., graph data extraction
 - We'll see several examples / techniques
- Simple application of LLMs to unstructured data is often not sufficient
- LLMs can work better with “smarter data”

← RDFox Blog

**SAMSUNG LAUNCHES GALAXY S25
SERIES WITH NEW AI FEATURES
BUILT ON TECHNOLOGY FROM
OXFORD SEMANTIC TECHNOLOGIES**

Graph data representation and
reasoning on mobile phones



Philip Foster
Chief Operating Officer

1 min read · January 23, 2025



Artificial General Intelligence vs Data Science

- Artificial General Intelligence

- How close is AI to human capabilities?

- AI for Data Science ... and in many other applications

- Data processing and analysis methods that are

- reliable (e.g., more reliable than humans and human memory)
 - efficient (e.g., batch processing)
 - useful for decision making (e.g.,

- Data that can be trusted

- Texts, images, videos only? What about key knowledge stored in structured or semi-structured sources (e.g., databases)?

Target Applications and Program

- Key target applications (data science & AI)
 - Semantic data integration and entity reconciliation
 - Information extraction and semantic data enrichment
 - Semantic search
 - Retrieval Augmented Generation (LLM + external sources)
- Program: lessons + hands-on
 - Introduction
 - Applications and semantics
 - Knowledge Graphs
 - Modelling languages, reasoning with ontologies, tools
 - Distributional semantics, Word Embeddings, LLMs
 - Transformers, learning representations, capabilities and risks
 - Construction of Knowledge Bases, Data Enrichment & Information Extraction

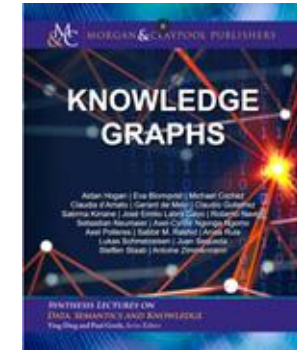
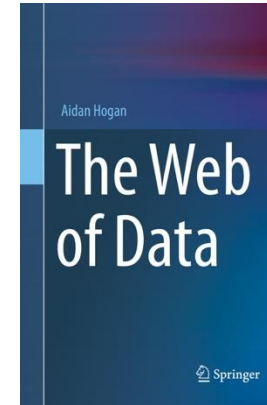
Why Knowledge Graphs

(Semantic Web)



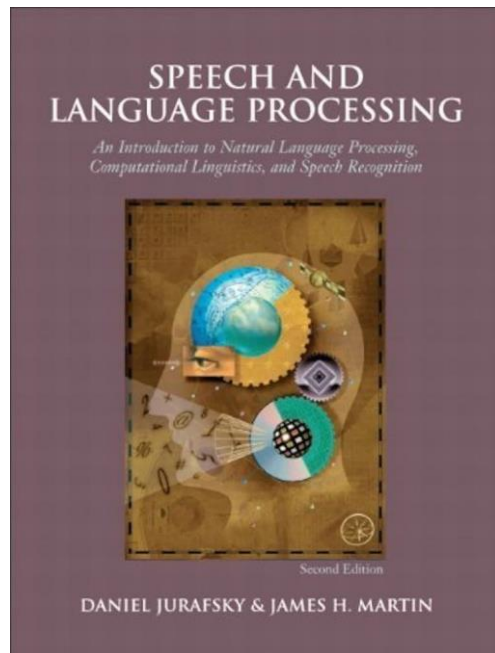
TEACHING MATERIAL

- **The Web of Data.** Aidan Hogan. 2020. Springer.
Pages 1-680. [Selected chapters]
- **Knowledge Graphs.** Aidan Hogan, Eva Blomqvist, Michael Cochez, Claudia d'Amato, Gerard de Melo, Claudio Gutierrez, Sabrina Kirrane, José Emilio Labra Gayo, Roberto Navigli, Sebastian Neumaier, Axel-Cyrille Ngonga Ngomo, Axel Polleres, Sabbir M. Rashid, Anisa Rula, Lukas Schmelzeisen, Juan Sequeda, Steffen Staab, and Antoine Zimmermann. Synthesis Lectures on Data, Semantics, and Knowledge, November 2021, Vol. 12, No. 2, Pages 1-257
(<https://doi.org/10.2200/S01125ED1V01Y202109DSK022>)
 - Free online version: <https://kgbook.org/#sec-ack>
 - Paper version: <https://arxiv.org/abs/2003.02320>
- Selected papers for second part



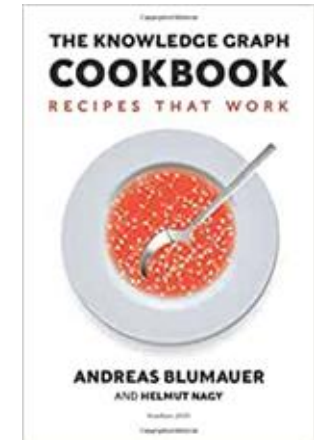
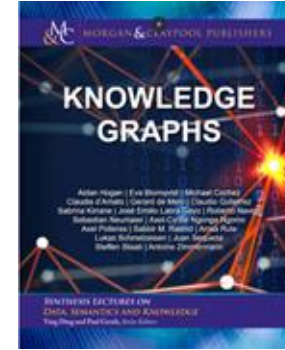
TEACHING MATERIAL

- Daniel Jurafsky and James H. Martin. 2025. **Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models**, 3rd edition.
 - Online manuscript released January 12, 2025.
<https://web.stanford.edu/~jurafsky/slp3>



TEACHING MATERIAL +

- Kejriwal, Mayank, Craig A. Knoblock, and Pedro Szekely. [*Knowledge Graphs: Fundamentals, Techniques, and Applications*](#). MIT Press, 2021.
- Blumauer, Andreas, Helmut Nagy. *The knowledge graph cookbook*. 2020. ([free download with questionnaire](#))



EVALUATION



- Aggregation of the scores obtained in two independent assessments.
- First assessment:
 - exam-tailored project (individuals or groups)
 - oral presentation supported by slides lasting about 20 minutes (with short demo of the project if any)
 - **project**: in-depth knowledge and/or hands-on experience of a specific topic covered in the course or linked to topics covered in the course;
 - Evaluated by: significance, methodological soundness, mastery of the in-depth topic.
- Second assessment:
 - **Assignments** to be completed before the oral exam, presented and evaluated at the exam